

Assignment 1 – Data Science Lab

Q1. Demonstrate the basic applications of NumPy arrays, Pandas DataFrames, and Matplotlib plots

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

def program1():
    # ----- NumPy Array Operations -----
    arr = np.random.randint(1,20, size=10)
    print(f"Original Array: {arr}")

    print(f"Mean of the given array: {np.mean(arr)}")
    print(f"Sum of the given array: {np.sum(arr)}")
    print(f"Sorted arr: {np.sort(arr)}")

    # ----- Pandas DataFrame Operations -----
    data = {
        'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Eva'],
        'Age': [24, 30, 22, 35, 29],
        'Score': [85, 90, 78, 88, 92]
    }
    df = pd.DataFrame(data)
    print("Data Frame :\n", df)

    print(f"Average age: {df['Age'].mean()}")
    print(f"Maximum Score : {df['Score'].max()}")
    print("Summary Statistics :\n", df.describe())

    # ----- Matplotlib Visualization -----
    plt.figure(figsize=(5,5))
    plt.bar(df['Name'], df['Score'], color='red', edgecolor="black")
    plt.xlabel("Students")
    plt.ylabel("Scores")
    plt.title("Scores of Students")
    plt.show()
```

Explanation: NumPy was used for generating random integer arrays and computing mean, sum, and sorting. Pandas DataFrame demonstrates data handling with average, maximum, and descriptive statistics. Matplotlib visualizes student scores using a bar chart.

Q2. Program for Frequency Distribution, Variability, and Averages

```
import numpy as np
import matplotlib.pyplot as plt
```

```

import pandas as pd

def program2():
    data = [12, 15, 14, 10, 18, 20, 15, 12, 17, 14,
            15, 16, 14, 18, 20, 19, 15, 12, 14, 16]
    series = pd.Series(data)

    # Frequency Distribution
    freq_dist = series.value_counts().sort_index()
    print("Frequency Distribution:\n", freq_dist)

    # Plot frequency distribution
    plt.figure(figsize=(7,5))
    freq_dist.plot(kind="bar", color="lightgreen", edgecolor="black")
    plt.title("Frequency Distribution")
    plt.xlabel("Value")
    plt.ylabel("Frequency")
    plt.show()

    # Averages
    mean_val = np.mean(data)
    median_val = np.median(data)
    mode_val = series.mode().values[0]

    print("\nAverages:")
    print("Mean:", mean_val)
    print("Median:", median_val)
    print("Mode:", mode_val)

    # Variability
    variance_val = np.var(data, ddof=1)
    std_dev_val = np.std(data, ddof=1)
    range_val = np.max(data) - np.min(data)

    print("\nVariability:")
    print("Variance:", variance_val)
    print("Standard Deviation:", std_dev_val)
    print("Range:", range_val)

    # Histogram
    plt.figure(figsize=(7,5))
    plt.hist(data, bins=6, color="skyblue", edgecolor="black")
    plt.title("Histogram of Data")
    plt.xlabel("Bins")
    plt.ylabel("Frequency")
    plt.show()

```

Explanation: Frequency distribution shows how often each value occurs in the dataset. Averages (mean, median, mode) provide measures of central tendency. Variability (variance, standard deviation, range) quantifies how data is spread out. Visualization includes bar plot and histogram.

Q3. Python Program for Normal Curves

```

import numpy as np
import matplotlib.pyplot as plt

```

```

from scipy.stats import norm

def program3():
    mean = 0
    std_dev = 1
    x = np.linspace(-4, 4, 1000)

    y = norm.pdf(x, mean, std_dev)

    # Normal Curve
    plt.figure(figsize=(8,5))
    plt.plot(x, y, color="blue", label=f"N({mean}, {std_dev}^2)")
    plt.fill_between(x, y, color="skyblue", alpha=0.4)
    plt.title("Normal Distribution Curve")
    plt.xlabel("X values")
    plt.ylabel("Probability Density")
    plt.legend()
    plt.grid(True)
    plt.show()

    # Multiple Curves with Different Std Devs
    std_devs = [0.5, 1, 2]
    plt.figure(figsize=(8,5))
    for sd in std_devs:
        y = norm.pdf(x, 0, sd)
        plt.plot(x, y, label=f"Mean=0, StdDev={sd}")

    plt.title("Comparison of Normal Curves")
    plt.xlabel("X values")
    plt.ylabel("Probability Density")
    plt.legend()
    plt.grid(True)
    plt.show()

```

Explanation: Normal distribution is a symmetric probability distribution defined by mean and standard deviation. The first plot shows a standard normal curve with mean 0 and standard deviation 1. The second plot compares curves with the same mean but different standard deviations, showing the effect of spread.